

FTML practical session 14

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1 LOWER BOUND ON THE PERFORMANCE OF GRADIENT-BASED ALGORITHMS

In previous sessions, we have discussed several upper bounds on the performance of machine learning methods.

For instance, we have seen that for a strongly convex smooth function f defined on $\mathbb{R}^d$ and with real values, the convergence of gradient descent to the minimizer x^* of f can be upper bounded. More formally, we have seen that if $(x^t)_{t \in \mathbb{N}}$ is the sequence of iterates, then

$$\|x^t - x^*\| = \mathcal{O}(\alpha^t) \quad (1)$$

with $0 \leq \alpha < 1$. If f is only convex, and not necessary strongly convex, then we have a weaker upper bound :

$$f(x^t) - f(x^*) = \mathcal{O}(1/t) \quad (2)$$

Having an upper bound guarantees that the algorithm converges **faster** than the bound.

In this session we present the problem of obtaining **lower** bounds on the performance of machine learning methods. This means proving that an algorithm can not be guaranteed to converge too fast to the minimizer of f . We will formally define what this means in section 1.3, as it is not as straightforward as for upper bounds.

There are two kinds of lower bounds : statistical lower bounds and optimization lower bounds, which we will focus on.

Let $\mathcal{F}$ be the space of functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that

- are convex and differentiable
- and have a minimizer x_f^*
- are L -smooth, with $L \in \mathbb{R}_+$ (the gradient is L -Lipshitz continuous).

1.1 First-order methods

We consider optimization algorithms that are based on linear combinations of local estimations of the gradient of f . These methods are called **first-order methods**. GD is an example of first-order method. Formally, this means that each iterate x^t verifies

$$x^t \in x^0 + \text{span}\{\nabla f(x^0), \dots, \nabla f(x^{t-1})\} \quad (3)$$

"span" means "espace engendré par".

1.2 Nesterov acceleration (AGD)

Nesterov acceleration (also called accelerated gradient descent (AGD)), is another first-order method, slightly different than GD. It is possible to prove that with Nesterov acceleration, the rate of convergence is $\mathcal{O}(1/t^2)$, instead of $\mathcal{O}(1/t)$.

<https://blogs.princeton.edu/imabandit/2013/04/01/acceleratedgradientdescent/>

1.3 Minimax lower bound

It is possible to show that AGD is optimal among first order methods. But first, we have to define what this means. This optimality is defined as a minmax problem. Indeed, we are interested in finding the algorithm that has the best worst-case performance. It is necessary to restrict algorithms to a relevant class of algorithms $\mathcal{A}$ (here, first order methods), and the functions to a relevant class of functions $\mathcal{F}$ (for instance the set of convex, L -smooth functions on $\mathbb{R}^d$).

For $t \in \mathbb{N}$, we look for an algorithm $a \in \mathcal{A}$ defined by

$$a = \arg \min_{a \in \mathcal{A}} \max_{f \in \mathcal{F}} \left[\|f(x_a^t) - f(x_f^*)\| \right] \quad (4)$$

where x_a^t is the iterate returned by algorithm a at iteration t .

1.4 Optimality of Nesterov acceleration

If we manage to show that for any algorithm $a \in \mathcal{A}$, there exists a function $f \in \mathcal{F}$, such that

$$f(x_a^t) - f(x_f^*) \geq \frac{C}{t^2} \|x^0 - x_f^*\|^2 \quad (5)$$

where C is independent on the function, then this will prove that Nesterov acceleration is optimal in $\mathcal{A}$, as this means that for any algorithm in $\mathcal{A}$, there exists a function, for which the iterates computed by the algorithm a converge slower than $\frac{C}{t^2}$ to its minimizer.

1.5 Hard function

Exercise 1 : Without loss of generality, we can assume that for all algorithms in $\mathcal{A}$, the initialization is $x^0 = 0 \in \mathbb{R}^d$: why?

We now consider that the initialization is $x^0 = 0$, $t \leq \frac{d-1}{2}$ and f defined by

$$f(x) = \frac{L}{4} \left(\frac{1}{2} x_1^2 + \frac{1}{2} \sum_{i=1}^{2t} (x_i - x_{i+1})^2 + \frac{1}{2} x_{2t+1}^2 - x_1 \right) \quad (6)$$

Exercise 2 : Show that f is convex.

We admit that f is L -smooth, and that the minimizer of f , x^* , is

$$x_i^* : \begin{cases} 1 - \frac{i}{2t+2} & \text{for } 1 \leq i \leq 2t+1 \\ 0 & \text{for } i \geq 2t+2 \end{cases}$$

and that the minimum value is

$$f^* = \frac{L}{8} \left(\frac{1}{2t+2} - 1 \right) \quad (7)$$

We consider another utility function g defined as

$$g(x) = \frac{L}{4} \left(\frac{1}{2} x_1^2 + \frac{1}{2} \sum_{i=1}^{t-1} (x_i - x_{i+1})^2 + \frac{1}{2} x_t^2 - x_1 \right) \quad (8)$$

We also admit that the minimum value of g , noted g^* , is

$$g^* = \frac{L}{8} \left(\frac{1}{t+1} - 1 \right) \quad (9)$$

Exercise 3: Compute the gradient of f for any point $x \in \mathbb{R}^d$.

Exercise 4: Show that for all $l \leq d$, the components j with $j \geq l+1$ are null for the iterate x^l , for any first-order method that produces iterates with gradients of f (as defined in 3). In particular, this is true for $l = t$ and for the iterate t , we have

$$x_{t+1}^t = x_{t+2}^t = \dots = x_d^t = 0 \quad (10)$$

Exercise 5: Show that $f(x^t) = g(x^t)$.

Hence, $f(x^t) \geq g^*$ and

$$\frac{f(x^t) - f^*}{\|x^0 - x^*\|^2} \geq \frac{g^* - f^*}{\|x^0 - x^*\|^2} = \frac{g^* - f^*}{\|x^*\|^2} \quad (11)$$

Exercise 6: Show that

$$\|x^*\|^2 \leq \frac{2t+2}{3} \quad (12)$$

Exercise 7: Conclude.

1.6 Discussion

By essence, considering the worst-case performance (the hardest function to optimize) is pessimistic. Some functions optimized with a first-order method converge faster than $\mathcal{O}(1/t^2)$. We have seen this behavior for a least-squares setting in a previous session.